

LECTURE 24: MATH1061/7861
TOPICS: RELATIONS ON SETS; REFLEXIVITY,
SYMMETRY, TRANSITIVITY
DATE: THURSDAY APRIL 23, 2026

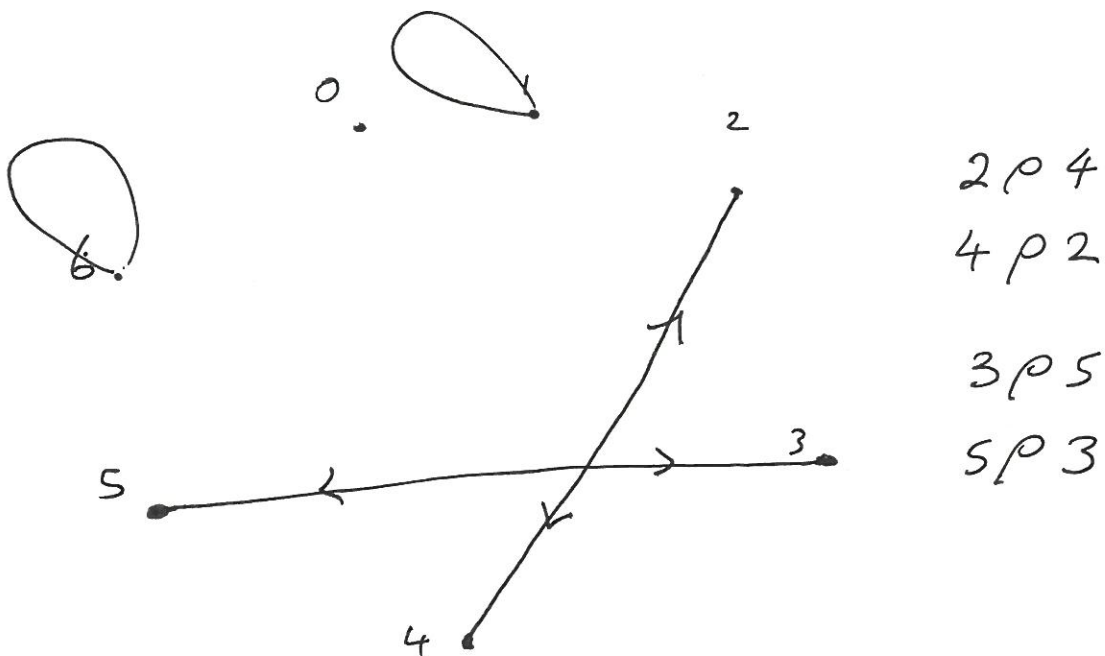
Learning objectives:

- (1) Understand binary relations on sets.
- (2) Learn how to draw an arrow diagram for a binary relation.
- (3) Learn the definitions of reflexivity, symmetry and transitivity for binary relations.
- (4) Learn how to prove or disprove these three properties.

Average on Exercise Sheets: ~ 87%.

Q1:

- (a) Draw an arrow diagram for the relation ρ on $\{0, 1, 2, 3, 4, 5, 6\}$ defined by $x\rho y$ if and only if $xy \equiv 1 \pmod{7}$.



set of all subsets.

↓

Q2: Let ρ be the relation on $\mathcal{P}(\mathbb{N})$ defined by $x\rho y$ if and only if $x \cap y = \emptyset$.

Determine if ρ is reflexive, symmetric, or transitive.

$$\{1, 2, 3\} \rho \{1000, 100000, \dots\}$$

Not Reflexive

$$\{1\} \not\rho \{1\}$$

$$\{1\} \cap \{1\} \neq \emptyset$$

✓ Symmetric

$$x \cap y = y \cap x$$

Not Transitive

$$X = \{1\} \quad Y = \{1000\} \quad Z = \{1\}$$

$$X \rho Y \text{ and } Y \rho Z \text{ but } X \not\rho Z.$$

(b) Determine whether this relation is reflexive, symmetric, or transitive.

✓ Symmetric Yes because $xy = yx$
 $xpy \Rightarrow ypx$ $\forall x, y \in \{0, 1, \dots, 6\}$

No Reflexive xpx $\forall x$ $2 \not p 2$ *

No Transitive ^{if} xpy and ypz then xpz
 $\forall x, y, z$

$$x = 2 \quad y = 4 \quad z = 2.$$

$$2p4, 4p2, 2 \not p 2$$

Q2

$\exists$ bijection $f: X \rightarrow Y$

(b) Define relation R on $\mathcal{P}(\mathbb{N})$ by

$x R y$ if $|x| = |y|$. $\leftarrow$ Same cardinality.

Determine if R is reflexive, symmetric, or transitive.

✓ Reflexive: $|X| = |X|$ $f: X \rightarrow X$
 $f(x) = x \quad \forall x \in X$.

✓ Symmetric: $|X| = |Y|$ means $f: X \rightarrow Y$ bijection
 $f^{-1}: Y \rightarrow X$ is also bijection so $|Y| = |X|$.

✓ Transitive $X R Y$ and $Y R Z$
 $f: X \rightarrow Y$ $g: Y \rightarrow Z$

$$\begin{array}{c} X \xrightarrow{f} Y \xrightarrow{g} Z \Rightarrow X R Z. \\ \quad \quad \quad \searrow \quad \quad \quad \nearrow \\ \quad \quad \quad g \circ f \end{array}$$

Exercise: If f and g are bijections then $g \circ f$ is also a bijection.